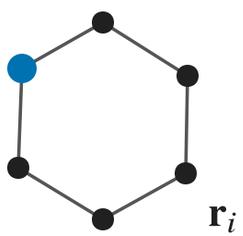
a Angular decomposition

$$u \otimes v = \underbrace{\tfrac{1}{3}(u \cdot v) I}_{L=0} + \underbrace{\tfrac{1}{2}(uv^{\mathsf{T}} - vu^{\mathsf{T}})}_{L=1} + \underbrace{\mathrm{STF}_2(u, v)}_{L=2}$$

b Coupled features and node update



$\mathbf{r}_i$

$\{Z_i, \mathbf{r}_i\}_{i=1}^{N}$

$Q_{ij} = \mathrm{STF}_2(\hat{r}_{ij}, \hat{r}_{ij})$

Clifford

$h_i^{\mathrm{Cl}} \in \mathrm{Cl}(3,0) \otimes \mathbb{R}^C$

$\mathrm{STF}_2(v_j, \hat{r}_{ij})$

$S_j \cdot v_j$

$S_i \in \mathrm{STF}_2 \otimes \mathbb{R}^C$

STF rank 2; rank 3 optional

$M_i = \sum_{j \in \mathcal{N}(i)} m_{j \to i}$

Augmented products

Gate · residual

Normalise

Repeat across layers

c Readout

Invariant scalar head

$$E = \sum_{t,i} \varepsilon_i^{(t)}$$

$$\varepsilon_i^{(t)} = \rho_t\left(\mathrm{Inv}(h_i^{(t)})\right)$$

Direct force sources

$$\langle h \rangle_1, \quad \star\langle h \rangle_2, \quad S\hat{r}$$

$$\downarrow$$

$$\mathbf{F}_i$$

$$\mathrm{STF}_2(u, v) = \tfrac{1}{2}(uv^{\mathsf{T}} + vu^{\mathsf{T}}) - \tfrac{1}{3}(u \cdot v) I$$